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The Challenge of Transportation Systems Analysis

1.1 A WORLD OF CHANGE

We live in a world of rapid change. This is particularly significant for transportation systems analysis because of the strong interactions between transportation and the rest of society.

We can identify three critical dimensions of change relevant to transportation. The first is change in the demand for transportation. As the population, income, and land-use patterns of metropolitan areas and states change, so do the patterns of demand for transportation—both the amount of transportation desired and the spatial and temporal distribution of that demand.

The second dimension of change is in technology. For example, in urban transportation, until just a few years ago the only actively considered alternatives were highways and rapid rail transit. Now we are able to consider such alternatives as lanes or even whole expressways restricted to buses; basically new technologies such as “dual-mode” systems, in which vehicles operate under individual control on local streets and automatically on tracked interurban guideways; and a variety of policy options designed to improve the efficiency of use of existing technology, such as incentives for carpools and van pools, “dial-a-ride” small buses, road pricing strategies, disincentives for automobile use, and auto-restricted zones (UMTA 1975, Smith, Maxfield, and Fromovitz 1977, TRANSPORTATION SYSTEM MANAGEMENT 1977). These new technologies provide a rich market basket of alternatives, from which a wide variety of transportation systems for metropolitan areas can be developed.

Change has been rapid in other areas of transportation technology as well, as exemplified by the development of freight containerization, “jumbo” jet aircraft, vertical or short takeoff and landing (V/STOL) aircraft, and air-cushion vehicles for water and land transport.

The third dimension of change is in the values, public and private, that are brought to bear on transportation decision making. It has become clear that many different groups are affected by decisions made about transportation. No longer is it sufficient to design transportation

systems simply to serve the “users,” in some aggregate sense. Rather, we must identify which groups are served well and which groups poorly by a particular facility or system; and so we have begun to focus on the needs of those who are too poor or too ill or too young or too old to have ready access to automobile transportation. We have also become deeply concerned with the social and environmental effects of transportation: air pollution, noise pollution, community disruption, and ecological effects are given increasing weight in transportation decision making.

These three dimensions of change—in demand, in technology, and in values—form the background against which we shall develop the basic concepts of transportation systems analysis.

1.2 THE SCOPE OF THE PROBLEM

The first step in formulating a systematic analysis of transportation systems is to examine the scope of the analytical task. We shall start by setting out the basic premises of our approach, namely, the explicit treatment of the total transportation system of a region and of the interrelations between transportation and its socioeconomic context. We shall then identify those aspects of the system that can be manipulated—the “options”—and those aspects that are relevant to decision making—the consequences, or “impacts,” of the options. Given this framework, we can proceed to discuss the problem of prediction (section 1.3).

1.2.1 Basic Premises

Two basic premises underlie our approach to the analysis of transportation systems:

1. The total transportation system of a region must be viewed as a single, multimodal system.
2. Consideration of the transportation system cannot be separated from consideration of the social, economic, and political system of the region.

THE TOTAL TRANSPORTATION SYSTEM

In approaching the analysis of a transportation systems problem, initially we must consider the total transportation system of the region:

1. All modes of transportation must be considered.
2. All elements of the transportation system must be considered: the persons and things being transported; the vehicles in which they are conveyed; and the network of facilities through which the vehicles,

passengers, and cargoes move, including terminals and transfer points as well as line-haul facilities.

3. All movements through the system must be considered, including passenger and goods flows from all origins to all destinations.

4. For each specific flow, the total trip, from point of origin to final destination, over all modes and facilities must be considered.

For example, in a study of intercity passenger transport in a megalopolitan region, initially we must consider railroads, airplanes, buses, private automobiles, and trucks as well as new and innovative modes such as tracked air-cushion vehicles (TACV) and V/STOL aircraft. We must consider not only the direct intercity line-haul links but also the vehicles that will operate over these links, the terminals, en route stations, and other transfer points, and such means for access to, and egress from, the intercity portion of the system as taxis, limousines, automobiles, local transit, and other means of intracity transport. We must consider the diverse patterns of origins and destination of movements as well as how passenger and goods flows may use the same facilities. In examining each movement, we must consider the service provided on access and egress portions of each trip as well as on the line-haul portion.

After this initial comprehensive definition of the transportation system has been made, the analyst, as he defines more finely the primary objectives of his analysis, can narrow his focus to those elements of the system that are of direct concern. This procedure will force him to consider explicitly the assumptions introduced by eliminating individual elements of a highly complex and interrelated system.

THE INTERRELATION OF TRANSPORTATION AND ACTIVITY SYSTEMS

The transportation system of a region is tightly interrelated with the socioeconomic system. Indeed, the transportation system will usually affect the way in which the socioeconomic system grows and changes. And changes in the socioeconomic system will in turn call forth changes in the transportation system. This interrelationship is fundamental to our view of transportation systems analysis.

The system of interest can be defined by three basic variables: **T**, the transportation system; **A**, the activity system, that is, the pattern of social and economic activities; and **F**, the pattern of flows in the transportation system, that is, the origins, destinations, routes, and volumes of goods and people moving through the system. Three kinds

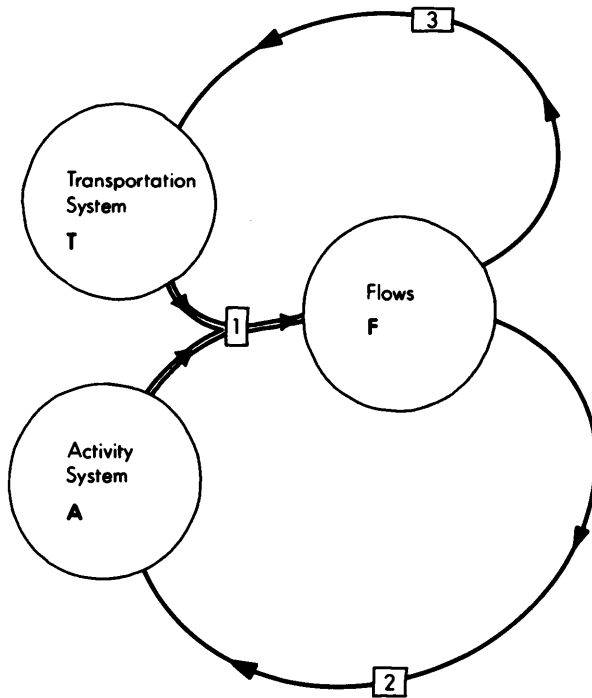


Figure 1.1 Basic relations.

of relationships can be identified among these variables (see figure 1.1):

1. The flow pattern in the transportation system is determined by both the transportation system and the activity system.
2. The current flow pattern will cause changes over time in the activity system: through the pattern of transportation services provided and through the resources consumed in providing that service.
3. The current flow pattern will also cause changes over time in the transportation system: in response to actual or anticipated flows, entrepreneurs and governments will develop new transportation services or modify existing services.

Though we label the activity system with the single symbol **A**, we must not assume that this system is as simple as the symbol suggests. On the contrary, the activity system of a metropolitan area or a megalopolitan region or a developing country consists of many subsystems, overlapping and interrelated—social structures, political institutions, housing markets, and so on. Transportation is only one of these subsystems.

The evolution of the activity system is determined by a large number of forces and pressures. The internal dynamics of this system are very complex, and our understanding of these dynamics is very incomplete.

Transportation plays a role in influencing the evolution of the activity system, but, except in very special situations, it is not the sole determinant of that evolution. The development of automobiles and of extensive systems of freeways does not alone cause suburbanization and dispersal of metropolitan areas, but it does interrelate closely with the dynamics of rising income, changing housing and labor markets, and other subsystems. Even the provision of access roads to a hitherto virgin area of an underdeveloped country will not by itself stimulate agricultural development. There must be a market for the produce, and there must be an array of adequate incentives to development.

The interrelation between transportation and the activity system is fundamental to our approach. ***The challenge of transportation systems analysis is to intervene, delicately and deliberately, in the complex fabric of a society to use transport effectively, in coordination with other public and private actions, to achieve the goals of that society.*** Responding to this challenge is not easy. We must understand transportation as a technology, a system of physical elements managed by human organizations to move people and goods. We must also understand transportation as a subsystem of the complex of social, economic, political, and other forces we so tersely summarize as "the activity system." Most important of all, we must know how to use this understanding effectively.

1.2.2 The Major Variables

Now that we have defined in broad terms the nature of the system with which we are dealing, we must explore the major variables needed in an analysis. In the last section we characterized the system in terms of three major, interrelated variables—T, A, and F. The questions we now address are: What options are available for influencing the system? What impacts should be considered in evaluating alternative courses of action?

INTERVENING IN THE SYSTEM: OPTIONS

There are many individuals, groups, and institutions whose decisions interact to affect the transportation system, the activity system, and thus the pattern of flows. The **user** of transportation, whether a shipper

of goods or a passenger, makes decisions about when, where, how, and whether to travel. The **operator** of particular transportation facilities or services makes decisions about vehicle routes and schedules, prices to be charged and services offered, the kinds and quantities of vehicles to be included in the fleet, the physical facilities to be provided, and so on. **Governments** make decisions on taxes, subsidies, and other financial matters that influence users and operators, on the provision of new or improved facilities, and on legal and administrative devices to influence, encourage, or constrain the decisions of operators or users.

It is often important to identify which groups have control over particular decisions, particularly when it is time to implement a selected course of action. We shall often ignore this question, however, in order to clarify the task of analysis. In other words, we shall attempt to identify all the possible decisions that might be made, without regard to who has the power to make a specific decision in a particular context.

Options, or decision variables, are those aspects of the transportation and activity systems that can be directly changed by the decisions of one or several individuals or institutions. The options available can be divided into two groups: those dealing with the transportation system itself and those dealing with the activity system.

Transportation options

Many aspects of a transportation system can be varied. Not all of these are open to a single decision maker, nor are all open at the same time. This spectrum of options, or "decision variables," may be summarized as follows:

Technology The development and implementation of new combinations of transportation components enable transportation demand to be satisfied in ways not previously available. Examples are containers, container ships, and piggyback trucks and railcars; the supersonic transport; and new urban mass transportation concepts, such as dual-mode and "dial-a-ride" systems.

Options involving technology include fundamental decisions about the means of propulsion, the medium through which the vehicle travels, supporting way and suspension systems, vehicle size and shape characteristics, typical route and network structure, and general mode of operations. Decisions must be made about these options within the constraints of technological feasibility, but there is a wide range of

options nevertheless, and we have seen very rapid growth in the variety of specific technologies available for urban, interurban, and developing country contexts.

Networks Options involving networks include their general configurations and the approximate geographical locations of their links. Examples are the grid systems typical of many of our present cities versus radial links and concentric circles.

Link characteristics Networks consist of links and nodes. Links correspond to facilities, such as highways, rail lines, or urban streets. Where it is necessary to model the characteristics of intersection points within a single mode (highway intersections, rail yards) and of transfer points between modes (airports, rail terminals, bus stops), these are also represented as links. Nodes simply express the connectivity relations of links in the network. Options include the detailed physical location of links and nodes and those characteristics of the links that affect flow, such as the number of lanes of highway or tracks of railroad, the grades and curves of the roadway, the type of signaling or traffic control, and the internal layout of terminals.

Generally we shall adopt the convention of most transportation network analysis and assume that all flow properties are represented in links and that nodes create no barriers to flow. Where node properties must be modeled—for example, transfer time at a rail terminal or airport—we shall do so by a subnetwork of links in the model that represents, and has the same properties as, the terminals. Thus nodes serve only to express the topology of the network.

Vehicles Most transportation modes involve vehicles (exceptions: pipelines, conveyors). The major options include the number of vehicles in the system and their characteristics. (Note that the choice of technologies sets a broad range to such options as networks, links, vehicles, and operating policies, but detailed decisions must still be made within the feasible range.)

System operating policies This set of options includes the full spectrum of decisions about how the transportation system is operated. The networks, links, and vehicles establish an envelope of possibilities; within that envelope a large variety of detailed operating decisions must be made. These options include vehicle routes and schedules, types of services to be offered, including services auxiliary to transportation (passenger meals, diversion and reconsignment privileges for freight), prices (both general pricing policy and specific pricing decisions), financing, subsidies, and taxing schemes, and regulatory decisions. Some of these operating policy options can be varied almost

on a day-to-day basis; others, such as pricing policy and regulatory decisions governing the entry of new carriers, may be unchanged for decades.

Organizational policies This set of options includes a wide variety of management, organizational, and institutional decisions. Within a single transportation organization, public or private, there are many detailed decisions about functional and geographic structure. Within a region there are decisions about how the transport sector should be organized, including the numbers and types of institutions, the functions to be assigned to each, the relative domains of responsibility, and the channels of communication, coordination, and control.

This set of transportation options fully defines the space of possible transportation plans and policies. However, these options are exercised not in a vacuum but in the context of a system of social and economic activities.

Activity-system options

The activity system is defined as the totality of social, economic, political, and other transactions taking place over space and time in a particular region. These transactions, both actual and potential, determine the demand for transportation; in turn, the levels and spatial patterns of those interactions are affected in part by the transportation services provided. Therefore, in modeling transportation systems, we must clearly identify those options in the activity system that will affect transportation demands.

Travel options These are the options open to every potential user of the transportation system: whether to make a trip at all, where to make it, when, and how—by what mode and route. These options apply to the individual traveler and to the shipper of freight. The decisions actually made by the shipper or traveler will be based in part upon the perceived characteristics of the transportation system and in part upon the actual and potential patterns of transactions in the activity system. The aggregate result of all the individual decisions about travel is expressed as the demand for transportation.

Other activity options Most of the social, economic, and political actors in the activity system have a wide range of options about how, when, and where they will conduct their activities. Over the long term these options profoundly influence the demand for transportation. For example, as major changes in a transportation system are made over time, the spatial pattern of population and economic activity will change, as actors exercise their options for changing the location or

scale of their activities. Forces within the economy external to the transportation system, such as national economic policy, rapid social change, housing subsidies, or mortgage policy, may impact on the spatial pattern of activity and thus affect the demand for transportation.

In many transportation analyses most of these activity options—for example, rate of economic growth, sectoral and regional patterns of growth, aggregate population—must be treated as exogenous, completely uncontrollable by the transportation analyst. The exercise of some of these options by various decision makers will be partially influenced by transportation; for example, transportation will affect the detailed distribution of population and employment within a region. Still other options are controllable to some extent in explicit coordination with transportation options—for example, the control of land use through zoning and land-development incentives.

Whether fully controllable or not, however, the full set of transportation and activity system options must be considered in any analysis.

THE CONSEQUENCES OF TRANSPORTATION: IMPACTS

When evaluating alternative transportation systems, one would like to consider all the relevant consequences. Any change in the transportation system can potentially affect a variety of groups and interests. **Impacts** are those aspects of the transportation and activity systems that should be considered in evaluating possible changes to the transportation system.

The prospective impacts can be broken down as follows, in terms of the groups on which the impacts fall:

1. **User impacts:** Impacts on travelers and shippers of goods. Users are differentiated by location within the region, by trip purpose, and by socioeconomic group. Examples: suburban resident commuting to central-city job; low-income non-car-owning resident of center city traveling to health facilities.
2. **Operator impacts:** Impacts on operators of the transport facilities and services. Differentiated by mode, by link, and by route. Examples: air carrier, trucker, highway maintenance agency, port authority, toll-bridge operator.
3. **Physical impacts:** Impacts caused by the “physical presence” of transport facilities or services affect many who are neither users nor operators. These groups can be differentiated by type of impact and by location. Examples: families, jobs, and taxable real estate displaced by new construction; neighbors affected by environmental degradation

through noise, fumes, air pollution, or groundwater changes.

4. *Functional impacts:* The impacts on the activity system as users change travel patterns in response to transport system changes. Differentiated by location within region and by type. Examples: changes in retail sales areas in suburban shopping centers and central business districts; changes in production costs; changes in land values.

5. *Governmental impacts:* Differentiated by location and by level of government or agency type. Examples: municipal, state, or federal agencies; citizen groups; elected representatives.

An essential characteristic of transportation is the differential incidence of its impacts. Some groups will gain from any transportation change; others may lose. Therefore, transportation choices are essentially sociopolitical choices: the interests of different groups must be balanced. This view has profound implications for the evaluation of alternative options.

1.3 PREDICTION OF FLOWS

Any proposed change in a transportation system (or a completely new system) can be expressed in terms of the options identified in section 1.2. The problem of prediction is to anticipate the impacts that a particular proposal will have; that is, we need procedures for predicting the impacts associated with any set of options (figure 1.2). In transportation the impacts depend upon the pattern of flows resulting from the particular set of options.

Consider the present transportation system **T** and activity system **A**. A particular proposed plan will be defined in terms of changes in the transportation options, ΔT , and in the activity-system options, ΔA . Implementation of the plan will change the transportation system from **T** to **T'** and the activity system from **A** to **A'**. Corresponding to these changes there will be a change in the pattern of flows: **F** will become **F'**.

The core of any transportation systems analysis is the prediction of changes in flows. There will usually be many other significant impacts as well, but predicting the change in flows is always an essential step. (Even if there is no change in flows, this judgment must be reached explicitly.)

Specification of the transportation system **T** at any point in time and of the activity system **A** implies the pattern of flows **F**. The basic hypothesis underlying this statement is that there is a market for transportation which can be separated out from other markets (Beckmann, McGuire,

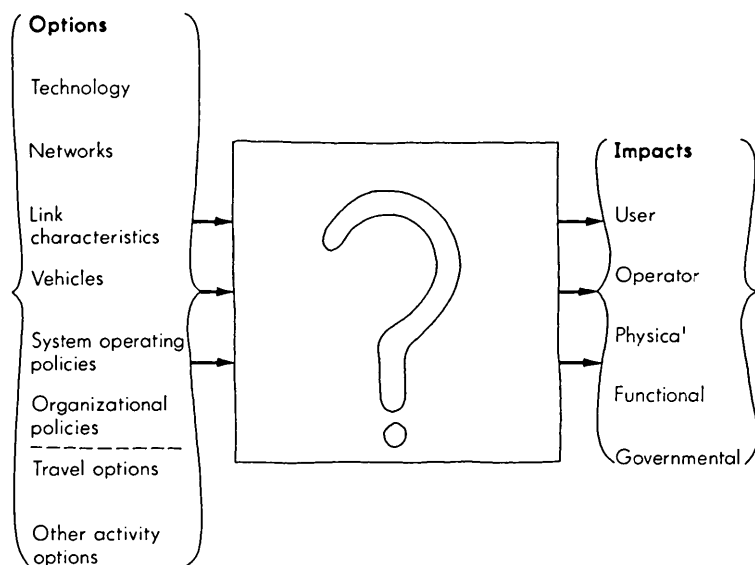


Figure 1.2 The prediction problem.

and Winston 1956, Manheim 1966b, Wohl and Martin 1967). This market is represented by the type 1 relation introduced in section 1.2.1; the hypothesis is that the type 1 relation can be separated from the type 2 and type 3 relations. This hypothesis can be expressed symbolically as follows. First, to our three variables **T**, **A**, and **F** we add two more: **S**, the service characteristics experienced by a particular flow or set of flows (travel times, fares, comfort, and so forth), and **V**, the volume(s) of flow in the network. Each of these variables may be a vector or other array.

We express the hypothesis as follows:

1. Specification of the transportation system **T** establishes **service functions, J**. These service functions indicate how the level of service varies as a function of the transportation options and the volume of flows; for a particular transportation system **T**, the level of service **S** that a traveler will experience is a function of the volume **V** of travelers using the system:

$$\mathbf{S} = \mathbf{J}(\mathbf{T}, \mathbf{V}). \quad (1.1)$$

2. Specification of the activity-system options, **A**, establishes **demand functions, D**. These demand functions give the volume of flows as a function of the activity-system options and the level of service; for a particular activity system **A**, the volume of travelers **V** that will use

the system is a function of the level of service **S** experienced by those travelers:

$$\mathbf{V} = \mathbf{D}(\mathbf{A}, \mathbf{S}). \quad (1.2)$$

3. The flow pattern **F** consists of the volume **V** using the system and the level of service **S** experienced by those travelers:

$$\mathbf{F} = (\mathbf{V}, \mathbf{S}). \quad (1.3)$$

For a particular transportation system **T** and activity system **A**, the flow pattern that will actually occur, $\mathbf{F}^0 = \mathbf{F}(\mathbf{T}, \mathbf{A})$, is the volume $\mathbf{V}^0$ and the level of service $\mathbf{S}^0$ determined as the equilibrium solution to the service and demand relations (1.1) and (1.2):

$$\left. \begin{array}{l} \mathbf{S} = \mathbf{J}(\mathbf{T}, \mathbf{V}) \\ \mathbf{V} = \mathbf{D}(\mathbf{A}, \mathbf{S}) \end{array} \right\} \longrightarrow (\mathbf{V}^0, \mathbf{S}^0). \quad (1.4)$$

Thus the specification of **T** and **A** implies particular values of equilibrium volume $\mathbf{V}^0$ and level of service $\mathbf{S}^0$ (if a unique equilibrium exists—see chapters 8 and 12):

$$(\mathbf{T}, \mathbf{A}) \longrightarrow (\mathbf{J}, \mathbf{D}) \longrightarrow [\mathbf{F}(\mathbf{T}, \mathbf{A}) = (\mathbf{V}^0, \mathbf{S}^0)]. \quad (1.5)$$

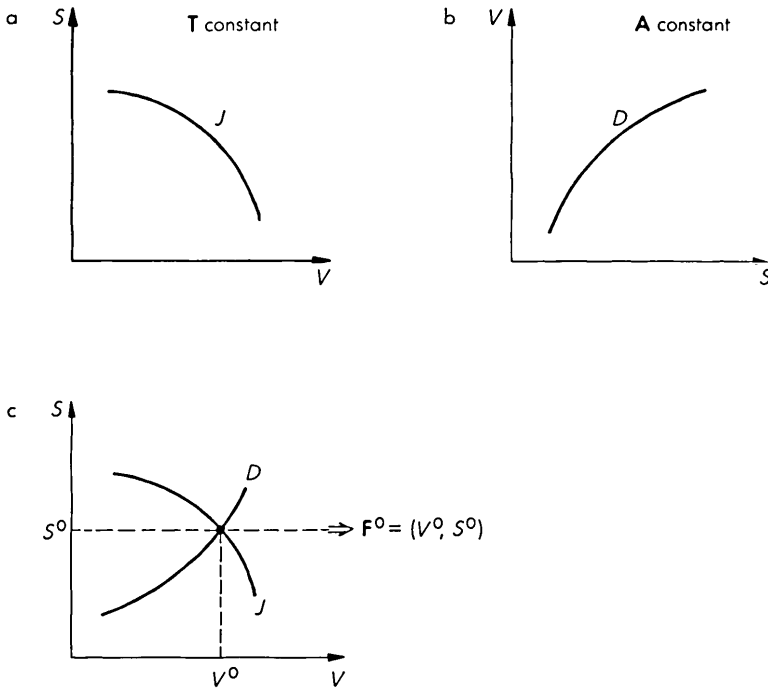


Figure 1.3 Simple equilibrium.

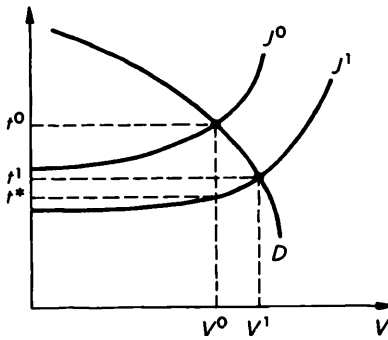


Figure 1.4 Comparing two systems.

The graphical interpretation of this formulation is shown in figure 1.3. In this figure V and S are assumed one-dimensional. Further, it is assumed that as the volume of flow in the system increases, the level of service decreases, as shown in part a of the figure; and as the level of service increases, the volume desiring to use the service increases, as shown in part b.¹

To see the implications of this formulation, suppose that we are considering two alternative transportation systems, T^0 and T^1 . T^0 is the existing transportation system, for example, a highway between two towns. We are considering replacing the highway by a new, improved facility T^1 . Figure 1.4 shows the two service functions, J^0 and J^1 , corresponding to T^0 and T^1 . Let us assume that there is one service attribute that is important: travel time over the routes, t . (Since improvement in S corresponds to reduction in t , the curves in this figure are opposite in shape to those of the previous figure.)

The equilibrium flow over T^0 is the flow $F^0 = (V^0, t^0)$ determined by the intersection of J^0 and D , the demand curve. Now consider the improved system T^1 , represented by J^1 . If we assume that the same volume of travel V^0 will occur on the new system as on the old, we

¹Note that the graphical conventions vary in the figure. In engineering the usual convention is that the independent variable is shown on the horizontal axis and the dependent variable on the vertical axis. This is followed in parts a and b. From the perspective of transportation system performance, V is the independent variable and S the dependent. From the perspective of the activity-system behavior, S is the independent variable and V the dependent variable, as expressed by equations 1.1 and 1.2. In part c the convention is arbitrary, since both variables are inter-related. We have chosen the economist's convention, in which V is on the horizontal axis and S (usually price in the economics literature) is on the vertical axis. The shape of the curves reflects the convention that "service" is positively valued.

would anticipate a service level t^* : that is, if volume remains constant, we expect a lower trip time because of the improved facility.

However, the constant-volume assumption is erroneous, for the travel volume will increase because the increased level of service—the decreased trip time—will attract more users. The extent of this increase in volume is given by the demand function D . Thus the actual flow pattern resulting will be that given by the equilibrium of D and J^1 : $F^1 = (V^1, t^1)$. That is, the traffic volume will increase, and the level of service will be intermediate between t^0 and t^* : the new facility will serve more users at a level of service that is better, but not as good as it would be if no new users were attracted.

1.4 APPLYING THE CONCEPTS

1.4.1 Simple Equilibrium: An Example

To illustrate these concepts we consider a highway connecting two towns, Suburb and City. We assume the following characteristics.

SERVICE LEVEL

The level of service S will be expressed by the travel time t for a trip between the two towns.

TRANSPORTATION SYSTEM

The road is a two-lane highway divided into two one-lane roadways, one in each direction. It is ten miles long.

SERVICE FUNCTION

We consider each of the roadways separately. The general form of the service function is

$$S = J(T, V) \quad (1.6a)$$

or, in this example,

$$t = m + nV, \quad (1.6b)$$

where options T are reflected in the values of the parameters m and n :

$$T = (m, n). \quad (1.7)$$

For this particular highway the parameter values are

$$\begin{aligned} m &= 10 \text{ minutes,} \\ n &= 0.01 \text{ minute per vehicle/hour.} \end{aligned} \quad (1.8)$$

That is,

$$t = 10 + 0.01V. \quad (1.9)$$

The units of t and V are thus minutes and vehicles per hour, respectively.

ACTIVITY SYSTEM

The two towns are characterized by their populations, employment levels, and income levels. The trip-making behavior of the residents reflects these variables.

DEMAND FUNCTION

We consider the one-way demand for travel from City to Suburb only. The general form of the demand function is

$$V = D(A, S) \quad (1.10a)$$

or

$$V = a + bt, \quad (1.10b)$$

where options A are reflected in the values of the parameters a and b :

$$A = (a, b). \quad (1.11)$$

For travel between the two towns the parameter values are

$$\begin{aligned} a &= 5,000 \text{ vehicles/hour,} \\ b &= -100 \text{ vehicles/hour per minute.} \end{aligned} \quad (1.12)$$

Thus

$$V = 5,000 - 100t. \quad (1.13)$$

FLOW PATTERN

The flow pattern F is defined by the volume V of travelers from City to Suburb, in vehicles per hour, and the level of service they experience, expressed by the travel time t in minutes:

$$F = (V, t). \quad (1.14)$$

EQUILIBRIUM

The equilibrium flow pattern (V^0, t^0) corresponding to the options (T, A) will be such that both service and demand relations are satisfied:

$$\begin{aligned}t^0 &= m + nV^0 = 10 + 0.01V^0 \\V^0 &= a + bt^0 = 5,000 - 100t^0.\end{aligned}\tag{1.15}$$

1.4.2 Exercises

INSTRUCTIONS FOR ANSWERING QUESTIONS

Many of the questions that follow are “self-checking” in that their answers immediately follow them. First cover the page with a sheet of paper. Then slide the paper down to the solid square that announces the answer. Now, keeping the answer covered by the paper, read the question. Think through your answer and jot it down on a piece of scratch paper (especially if you are asked to sketch or calculate something). Then uncover the printed answer and compare with your solution. If correct, go on to the next question. If incorrect, go back and review the material until you understand why your answer was wrong.

■ **Question 1.1** Examine the values of the parameters a , b , m , and n as given in equations (1.8) and (1.12). Discuss briefly the physical significance of these values. What is the significance of their signs (+ or –)? of their relative magnitudes?

■ **Answer 1.1** *Parameters a and b :* The demand function describes the number of people or vehicles that will travel at different levels of service (considering travel time as the level of service). The parameter a can be considered as a potential demand, that is, the demand for travel if travel time between the two zones were zero. The equilibrium volume of flow will certainly be less than a . The parameter b represents the change in demand for each unit change of travel time in minutes. Note that $b < 0$, correctly indicating that as travel time increases, demand decreases.

Parameters m and n : The service function describes the level of service that a particular transportation option will provide for various flow volumes. The parameter m measures the free-flow travel time, the travel time over the link for zero volume. The parameter n represents the effects of congestion on travel time. Specifically, for each vehicle on the link, travel time increases by 0.01 minute per mile. Note that $n > 0$ correctly indicates that as V increases, t increases.

■ **Question 1.2** Find the equilibrium flow pattern graphically. Plot the service and demand functions on the same set of axes. (Place t on the horizontal axis, V on the vertical.) Determine the equilibrium

flow pattern F^1 from the intersection of the two functions. (The answer for this question follows the next question.)

■ **Question 1.3** Find the equilibrium flow pattern by solving the equations for F^1 algebraically. Check to see that your results match the graphical solution. (Remember that F^1 is determined by specifying both an equilibrium flow volume and a travel time.)

■ **Answers 1.2 and 1.3** The demand equation is

$$V = 5,000 - 100t,$$

and the service equation is

$$t = 10 + 0.01V.$$

Solving these simultaneous equations, we obtain

$$V = 2,000 \text{ vehicles/hour,}$$

$$t = 30 \text{ minutes.}$$

Thus

$$F^1 = (V, t) = (2,000, 30).$$

This is shown graphically in figure 1.5.

■ **Question 1.4** The highway department is considering building an alternative link connecting zones 1 and 2. This link would be characterized by the following service function:

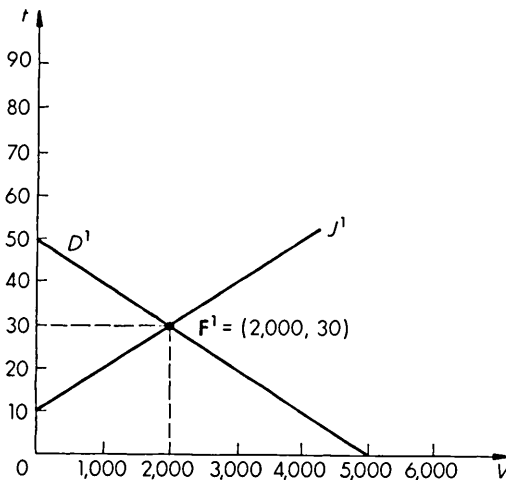


Figure 1.5 Graphical calculation of an equilibrium flow pattern.

$$t = 10 + 0.005V.$$

Plot this function on the same set of axes used in 1.2. Find the new equilibrium volume and travel time, graphically and algebraically. Discuss your results.

■ **Answer 1.4** The new equilibrium flow pattern is $F^2 = (2,666, 23.33)$. Notice that the new link results in decreased travel time and increased volume. Thus the improved facility has “induced,” or “generated,” $2,666 - 2,000 = 666$ new users on the link. The result is shown graphically in figure 1.6.

1.5 OTHER ELEMENTS OF PREDICTION

While the prediction of flows is an important part of the prediction of transportation impacts, it is not the whole of it. In section 1.2.1 we identified three major types of interrelations among the basic variables **T**, **A**, and **F**. The first relationship is that in which **T** and **A** determine **F**. As we have just seen, this type 1 relation corresponds to the hypothesis that there is a transportation market in which service and demand reach equilibrium, thus establishing the flow pattern $F = (V, S)$.

In addition to the type 1 relation, there are also type 2 and type 3 relations to consider. While the prediction of flow patterns is the core of an analysis, it is rarely the sole element of prediction; many other impacts must usually be predicted as well.

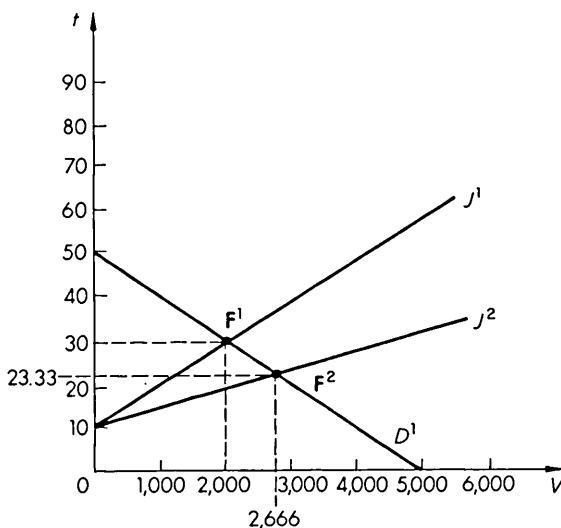


Figure 1.6 Effect of a change in service function on an equilibrium flow pattern.

1.5.1 Activity Shifts

We now explore the type 2 relationship: the effect of the current flow pattern in causing, over time, changes in the activity system. To do this we continue the simple graphical example of section 1.3.

Since it takes time to implement transportation system improvements while population and travel time continue to increase, the demand curve D^0 may shift upward and to the right, yielding a curve D^2 , as shown in figure 1.7. The corresponding equilibrium flow F^2 may be such that the improvement in trip time is even less.

This shift in the demand curve is, we assume, independent of any change in the transportation system. The forces influencing this shift are external, or exogenous, events such as population growth or changes in the economy. We designate these as E , for exogenous events; the shift in the demand curve will be a function of E .

The implementation of T^1 may cause further shifts in demand—for example, the development of residential subdivisions and shopping centers may follow the construction of a new highway, or the development of an industrial city may be a consequence of major rail or port development. Thus the demand curve shifts to D^3 . (The new equilibrium (V^3, t^3) may actually be such that t^3 is greater than t^0 : the level of service over the new system is actually worse than the level of service over the previous system at the initial period. However, it is better than the level of service t^{2*} that would have resulted from the old system T^0 and the shifted demand function D^2 .)

This effect takes place, of course, in the context of exogenous

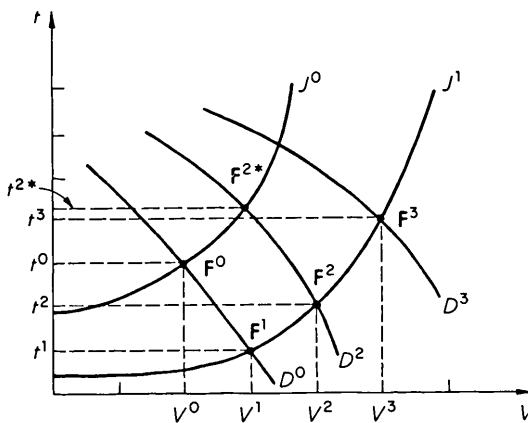


Figure 1.7 Activity shifts.

events E. We can represent this chain of effects schematically (t'' being later than t'):

$$(T', A') \longrightarrow [F' = (V', S')] \longrightarrow A''.$$

We call this the activity-shift relation: the pattern of service at time t' modifies the activity system at time t'' . From the point of view of a transportation analysis, the significance is that the demand curve at t'' is shifted from that at t' .

It is often useful to distinguish between these two relationships in the following manner. The type 1 relation involves short-run equilibrium—the pattern of flows that results from a particular activity system. The type 2 relation involves long-run equilibrium—shifts in the activity system itself in response to the flow pattern. We therefore call the type 1 relation **travel-market equilibration** and the type 2 relation **activity-system equilibration**.

1.5.2 Resource Consumption

Transportation systems consume resources such as energy, labor, materials and supplies, and land. Even if no change is made to a system, resources are consumed simply to provide service. If changes are made, especially involving major acquisitions of new vehicles or new guideway facilities, then the resources consumed can be quite substantial. Thus, in addition to predicting flows, we should also predict the resources consumed in providing and operating a particular transportation system.

1.5.3 The Operator Decision Loop

The resources consumed are of interest to a variety of groups, but especially to operators of transportation systems. Here we are concerned with how the operators of particular components of the transportation system decide when and how to adjust the transportation options with which they are concerned.

The structure of this decision is represented by the type 3 relation: the way the actual flows influence decisions concerning transportation options. We do not want to assume away the logic of that decision of what options to implement and thus of what level of service to offer. Rather, we want to isolate the decision variables that are within the control of the operator and include them explicitly in our vector of options. Thus a carrier may determine fares and schedules internally under the criterion of maximizing net revenue, for example. However, a

systems analyst would not want to assume that criterion, but would prefer to vary these options—schedules and fares—explicitly. That is, the analyst should deal with the full vector of options explicitly, regardless of which actors have control over particular subsets of options. Then he is in a position to try to identify those options that are within the control of the particular agency or firm for which he is doing the analysis, and to try to anticipate how other options will be manipulated by the actors who have control over them. Since the decisions of other actors may be influenced directly or indirectly, it is important that the decision-making logic of the various actors not be subsumed into the analyst's predictive models.

Therefore, in our approach to analysis we choose to keep all of the transportation decision options external to the prediction process. That is, we “break the loop” on the type 3 relation and do not model it explicitly, as we do with the type 1 and type 2 relations.

Of course, there will be some contexts in which it may be useful and desirable to model the operator's decision processes—for example, when an analyst working for one carrier attempts to develop a strategy in the face of competition from other carriers, or when a regulatory agency attempts to influence, through promotion or restriction, the actions of all carriers. In such cases appropriate models can be developed to predict the decisions of specific transport operators (see section 5.7, where we show how assuming an operator decision logic can lead to development of a “supply” function).

1.5.4 Systems of Models

Thus, to predict all significant impacts, five major types of models are required:

1. ***Service models*** are needed to determine, for any specified set of options, the levels of service at various flow volumes. Examples: travel time over a rail link as a function of train length, schedule, roadway conditions, and volume of passengers; volume vs. travel time curves as used in traffic assignment procedures.
2. ***Resource models*** are needed to determine the resources consumed (land, labor, capital, and other direct costs; air, noise, and other environmental impacts; aesthetic and social impacts) in providing a particular level of service with specified options.
3. ***Demand models*** are needed to determine the volume of travel demanded, and its composition, at various levels of service.
4. ***Equilibrium models*** are needed to predict the volumes that will

actually flow in a transportation system for a particular set of service and demand functions (short-term equilibrium in the travel market).

5. **Activity-shift models** are needed to predict the long-term changes in the spatial distribution and structure of the activity system as a consequence of the short-run equilibrium pattern of flows, that is, the feedback effect of transportation on land use (activity-system equilibrium).

These are the five basic components of any system of prediction models in transportation. The interrelationships among them are illustrated in figure 1.8. In addition, it would be desirable also to have models for predicting the changes in organizational and institutional behavior that result from changes in organizational policies. These models, however, are beyond the scope of this volume (see, for example, Allison 1971).

This structuring of the transportation systems analysis problem incorporates five hypotheses. The first hypothesis is that this is a complete and useful summary of the types of options and impacts. The second hypothesis is that it is meaningful to model transportation technology from two perspectives: in terms of the service perceived by prospective users, reflected in the service functions, and in terms of the resources consumed in providing that transportation service, reflected in the resource functions. The third hypothesis is that it is useful to separate short-term and long-term equilibrium: the short-term re-

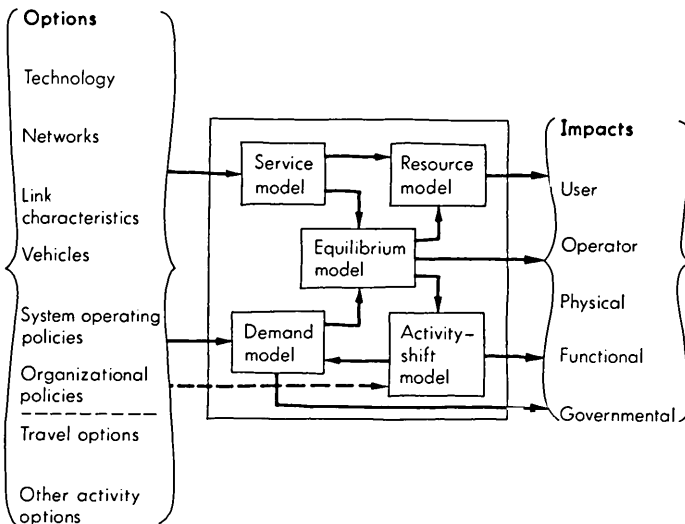


Figure 1.8 Basic prediction models.

sponses of transportation users, in a “transportation market” with the activity system fixed, as represented by the demand functions (the type 1 relation); and the long-term responses of users and others in a larger, more general market, the total economy, as represented by the activity shifts (the type 2 relation). The fourth hypothesis is that a unique equilibrium exists in each of these markets and that it can be found. The fifth hypothesis, which in a sense is the operational test of the second, third, and fourth hypotheses, is that valid predictive models can indeed be constructed.

1.5.5 The Variety of Applications (Optional Reading)

This framework has been applied (implicitly if not explicitly) in a wide variety of analytical activities, including urban transportation planning, intercity passenger transport studies, and national transportation planning in developing countries.

Urban transportation studies began in Detroit and Chicago in the mid-1950s. The prediction portion of the conventional urban transportation planning process consists of variants of the following sequence (Martin, Memmott, and Bone 1961, Hutchinson 1974):

1. Project land use, population, and employment changes.
2. Predict trip ends generated in each zone.
3. Predict interzonal distribution of trip ends (using gravity or opportunity models).
4. Predict modal split.
5. Predict distribution of flows over the proposed network.

These steps implicitly represent a travel-market equilibration (see chapter 11).

There are serious internal inconsistencies in this sequence, from the point of view of an equilibrium analysis (Wohl and Martin 1967, Manheim 1970b, 1973b). For example, the estimation of trip ends assumes implicitly a general level of service in the system, whereas the interzonal distribution calculations require an explicit level of service (derived by means of a gravity model, for example). The last step of the process, traffic assignment, predicts an “actual” level of service, or set of travel times, for flows in the network. However, the initial estimates of level of service used for trip generation and distribution are rarely revised to be consistent with the travel times predicted by the traffic assignment.

In spite of these inconsistencies and other limitations, the structure implicit in conventional urban transportation models is fundamentally

that described here. The service functions are represented as volume vs. travel time functions or simply link capacities and travel times. The demand functions are represented by predicting trip ends, interzonal distribution of trips, and then modal split. The travel-market equilibrium model is the "traffic-assignment" process, with the various "capacity-restraint" formulations representing explicit attempts to find equilibrium in the network, given fixed demands. (All-or-nothing assignments are obviously very difficult to justify as a meaningful prediction of "equilibrium" flows.) The resource models are represented in a variety of ad hoc calculations involving such factors as rights-of-way and construction and operating costs. Activity-shift models are sometimes explicit, as when land-use models are used to predict the effects of differential changes in accessibilities on the location of population and economic activities. More often they are left unstated.

While the conventional urban transportation planning models have serious limitations, a new generation of models is now being developed. These models encompass much improved demand functions and a sounder theoretical basis for explicit travel-market equilibrium analysis (see chapter 11).

In intercity passenger transport studies, this equilibrium structure has been more explicit. For example, in the Northeast Corridor Transportation Project—the first major intercity study—the system of models used included explicit demand models for passengers and freight (see section 4.3.2); technology models to produce service and resource functions; a network simulator for travel-market equilibration; and activity-shift models for forecasting changes in inter- and intraregional location and intensities of economic activities as a function of changes in transportation and other factors (Bruck, Manheim, and Shuldiner 1967).

The Harvard-Brookings Model System was designed for use in planning investment in transportation in developing countries (Kresge and Roberts 1971). Several explicit technology models were used for predicting service levels and resource consumption of highways, railways, and other modes and of intermodal transfer points. Demand for travel was derived from a macroeconomic model containing an inter-regional input-output model; these were also used to predict activity shifts. Network equilibrium was found with a modified traffic-assignment approach.

These examples illustrate some of the applications of the basic concepts. They are very cursory descriptions of highly sophisticated

systems of models, and they simply point out how the basic framework outlined above has been applied, implicitly or explicitly, in several major transportation analyses. In each application different practical approximations have been made.

1.5.6 Two Styles of Prediction

The prediction of the future impacts of alternative transportation plans is a difficult task. For plans that involve small changes to an existing system, and impacts in the near future, information about the present system and its effects can play a strong role. In this case it is useful to predict impacts by predicting the magnitudes of the changes from present conditions, leading to a style that we shall term ***incremental prediction***.

In incremental prediction the analyst knows the present states of the transportation and activity systems— T^0 , A^0 —the flow patterns— $F^0 = (V^0, S^0)$ —and the resources being consumed— R^0 . For any specified change ΔT^i in the transportation system, the impacts to be predicted are the incremental changes in the various elements— ΔV^i , ΔS^i , ΔR^i , ΔA^i . (These symbols generally represent vectors and may represent variations over time.)

In ***synthetic prediction***, or simply “prediction,” the analyst predicts the new values of V^i , S^i , R^i , and A^i directly.

The major difference is a practical one. Incremental prediction opens the way for application of powerful approximation methods. It is also good discipline for the analyst’s judgments about the magnitudes of changes likely to occur from existing conditions.

In later chapters we shall often show how short-cut, simplified analyses can be done using incremental instead of synthetic prediction.

1.5.7 Conclusions: The Elements of Prediction

The same basic theory is applicable to every transportation systems problem. Prediction of the impacts of the range of alternative transport plans requires, in general, the whole system of five models for service, demand, equilibrium, resources, and activity shifts. In actually applying the theory to a particular context, however, very different practical methods may be appropriate in different problems.

The principles presented in section 1.2 and the approach to prediction presented in this section are guidelines, not rigid prescriptions. One element of the art of doing a valid yet pragmatic and relevant analysis is to apply these concepts judiciously. The development of a long-range multimodal plan for an urban region might utilize a set

of approximations very different from that of an analysis of next month's schedule changes for a single bus route or the planning of a rail system in a developing country. The practical application of these concepts in a particular context requires understanding of the theoretical basis of the field and of the practical methods available in a specific situation.

This frames *the methodological challenge of transportation systems analysis*, which is to conduct an analysis in a particular situation which is valid, practical, and relevant.

1.5.8 Example (continued)

We continue with the example of section 1.4.

■ **Question 1.5** Often, while new transportation facilities are being planned and constructed, changes occur within the activity system. In our particular case, over the ten-year period of planning and construction of the new link described in question 1.1, the following changes occur in the activity system:

1. Population and job levels in the two zones increase.
2. On the average, people now place a relatively higher value on time.

Assume that the parameters of the new demand function describing the activity system at this future date are

$$a' = 7,500 \text{ vehicles/hour,}$$

$$b' = -150 \text{ vehicles/hour per minute.}$$

i. Compare these values with those given earlier in this section, for the existing conditions. Are they consistent with the activity-system changes described above? Briefly discuss your reasoning.

ii. Plot the new demand function on the same set of axes as in questions 1.2 and 1.4. Find the new equilibrium graphically or algebraically. Remember which service and demand functions we are now considering.

iii. What would the travel time over the existing link be if the new one had not been built?

■ **Answer 1.5**

- i. The two demand functions are

$$V = 5,000 - 100t,$$

$$V = 7,500 - 150t.$$

The demand parameters of the second function are consistent with the described changes within the activity system. Increases in population

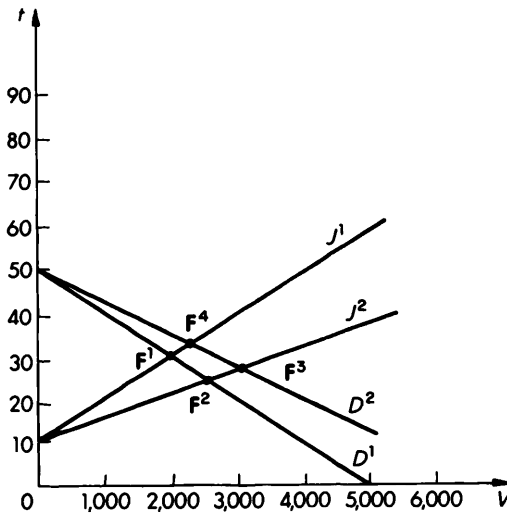


Figure 1.9 Effect of activity-system changes.

and job levels tend to increase the potential demand as expressed by parameter a' . As time is valued more dearly, the unit change in demand (for each unit of travel time) will increase. Thus b , which measures the sensitivity of demand with respect to travel time, becomes more negative as people value time more dearly.

ii. Solving algebraically:

$$\text{Demand: } V = 7,500 - 150t,$$

$$\text{Supply: } t = 10 + 0.005V.$$

Thus

$$V = 3,430 \text{ vehicles/hour,}$$

$$t = 27.15 \text{ minutes,}$$

$$F_3 = (V, t) = (3,430, 27.15).$$

Comparing this result to those of question 1.1, we see that both travel time and volume have increased. Referring to figure 1.9, we see that our equilibrium point has moved along the service curve because of the shift in demand.

iii. From the plot or algebra,

$$F_4 = (2,400, 34).$$

■ **Question 1.6** There are many examples of facilities being outdated from their very first day of service. For example, the Long Island

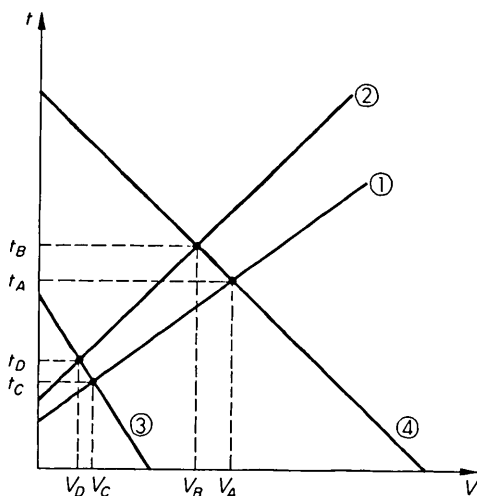


Figure 1.10 Effect of the introduction of a new highway.

Expressway was congested soon after it was opened (Caro 1974, p. 949). In fact, it is observed that travel times can increase on supposedly new, improved facilities. Discuss this phenomenon with reference to figure 1.10.

- i. Specifically, identify the functions that represent the old facility (highway), the new, improved facility, the old activity system, and the new activity system.
- ii. Describe the relative magnitudes of the equilibrium volumes and travel times for the old and new systems.
- iii. An irate traveler was quoted as saying, "Look, since they've built the new highway my trip time is worse than ever. I haven't derived any benefit from this road." Study the figure. Do you agree that the new road has made things worse for society? Ponder seriously, but comment only briefly.

■ **Answer 1.6**

- i. The old facility is represented by function 2, the new facility by function 1, the old activity system by function 3, and the new activity system by function 4.
- ii. Referring to the figure, it can be seen that the new equilibrium flow is (V_A, t_A) . Both travel time and volume are greater than the old equilibrium flow pattern (V_D, t_D) .
- iii. As paradoxical as it may seem, it must be concluded that users have benefited from the new system. Consider that most trips are made

for useful purposes (on urban highways, for example, we find that most trips are for work, shopping, school, or recreation). If a new system induces greatly increased travel, we can only conclude that more people are traveling to work, shopping centers, schools, and social functions. From the viewpoint of users, we see that more people are making more trips and thus deriving more benefits from the new highway than from the old. The increased travel time is an unfortunate by-product of urban growth. Of course, key questions are: What costs were incurred in order to build this new facility? What would have occurred if the facility was not built? Who benefited and who was hurt by the new facility?

It should be stressed that travel demand and changes in land use are not independent of changes in transportation systems. Often new highways spur tremendous growth in residential and industrial land use. For example, the Long Island Expressway brought about a tremendous growth in residential areas of Long Island. And Route 128 in the Boston area precipitated an industrial migration as large firms found it more profitable to locate in suburban areas (served by the new road) than in a downtown business district. Thus new highways can create jobs and spur residential land development. These benefits are often difficult to measure directly but can have great effects on society.

1.6 PUTTING PREDICTION IN CONTEXT

So far, in discussing options, impacts, equilibrium analysis, and model systems, we have been relatively abstract. What implications can we draw from this discussion? What steps should we go through in order to analyze a particular transportation system problem?

1.6.1 The Analysis Cycle

We start with a particular transportation system plan. This plan proposes certain transportation and activity systems described as sets of options. To assess the impacts of this plan, we must predict the flow pattern that will occur if it is implemented, that is, the short-run equilibrium of the transportation market and the long-run equilibrium effects of transportation on the location and character of social and economic activity. These predictions require a system of five models describing service, demand, equilibrium, resources, and activity shifts.

Thus the core of the transportation systems analysis problem is prediction of the impacts of a particular plan using a system of equilibrium models. **Prediction** is concerned with anticipating the consequences of alternative plans: What impacts will each alternative have? On

whom will they fall? For a particular pattern of available transportation services, what will be the flows? How will each alternative affect the evolution of the activity system?

But how did we get the plans in the first place? There must have been some kind of process by which a variety of possibilities were quickly examined and discarded, until some reasonably good candidates emerged. This is the **search process**—the generation of alternatives sufficiently well-defined to be tested in prediction. The search process may be algorithmic (a linear-programming procedure, for example) or completely intuitive (an analyst proposing a change in services or prices, or a designer sketching a network on a map) or some mixture of the two extremes. The questions one must ask in this process are: What possibilities for intervention are there? What aspects of the transportation system might be modified or improved? What alternative actions should we examine? What actions are most likely to achieve the objectives desired?

Once one or several alternative plans have been generated through some search procedure, and the impacts have been predicted using the system of prediction models, the next task is to evaluate those impacts, compare the predicted impacts for the various alternatives, and arrive at a judgment as to which alternatives are preferred, and why. **Evaluation** and **choice** are the activities through which the predicted consequences of alternative actions are distilled and weighed, to reach a decision on a course of action. They involve such questions as the following: What impacts would be most desirable? Which of the al-

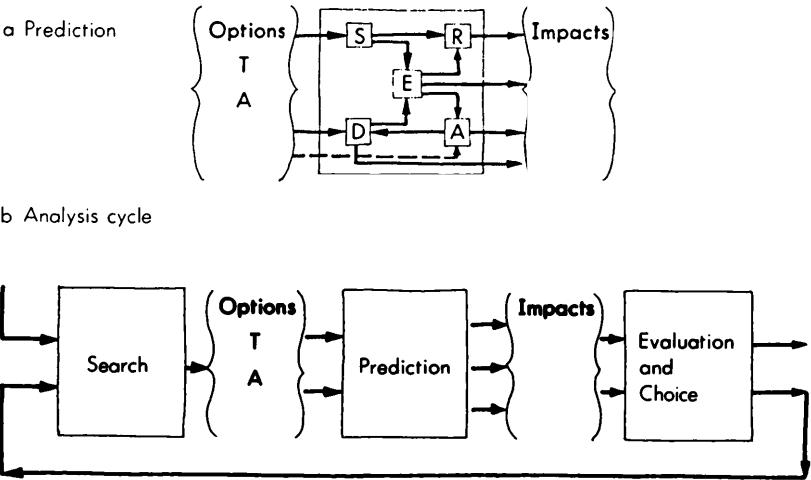


Figure 1.11 Prediction in the larger process.

ternative actions would be most desirable? Whose points of view should be considered in reaching a decision? What should be the process of reaching a decision? Which alternative should be implemented?

Figure 1.11 brings out the interrelations of these activities. Part b of the figure emphasizes the iterative nature of the process: after evaluation and choice, there should be the option of recycling to search again, one or more times, generating and evaluating a succession of alternatives. This iterative process may proceed until a desirable alternative is found or until the available resources (dollars, time) are exhausted.

1.6.2 Analysis and Implementation

Of course, even this view of the process of analysis is limited, and we must embed it in a yet larger picture. This is shown in figure 1.12, in which several additional distinctions have been made:

1. A **setup phase** precedes the technical analysis. In order to do the prediction, the system of models must be developed: data must be collected, initial goals formulated, and substantial development activity

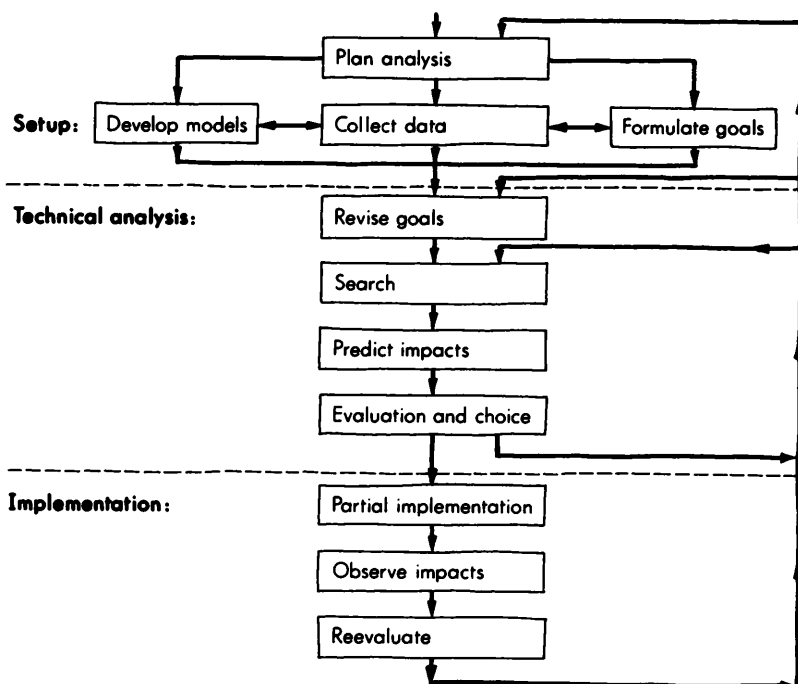


Figure 1.12 The continuing analysis process—Image 1.

must occur in order to produce appropriate service, demand, and other models. The analysis and setup phases must be planned.

2. The ***technical-analysis phase*** includes not only search, prediction, and evaluation and choice, but also a goal formulation and revision task. Initial statements of objectives must be formulated in the setup phase to provide guidance for the development of models and data collection. At the start of analysis these goals are revised to provide guidance for search and a basis for evaluation and choice. However, it is to be expected that, as the transportation analysts learn about the problem by iterating through search, prediction, evaluation, and choice, goals will be revised and refined (Manheim 1970a).

3. Transportation systems analysis is rarely conducted in a vacuum. There is usually a purpose to the analysis: to come to a conclusion as to a desirable course of action to be implemented in the real world. Therefore, we show an ***implementation phase*** following the technical analysis.

The central idea behind this figure is that transportation systems analysis is a dynamic activity. For existing systems, analysis of alternative options is a continuing task. For systems being expanded or built new, it must be recognized that change takes time. It may take many years to implement fully a particular course of action—building a highway or an airport, restructuring the routes and services of a transit system or an airline, building a major new freeway or transit system—even after it has been adopted in principle.

Many transportation analyses are still being set up as “one-shot” activities: do a study, recommend a course of action, and disappear. However, it is now becoming more widely recognized that a truly effective transportation systems analysis must be a continuing activity because of the long time period for implementation. As implementation of the recommended action gets under way, the actual impacts should be observed and compared with the predictions. Simultaneously, changes in technology, in demand, and in values should also be monitored. Based upon this new information, the analysis and even setup phases should be recycled, to adapt the previously recommended action to changing conditions.

1.6.3 The Product of Analysis

To emphasize the continuous nature of the task, the relation of analysis and implementation might be visualized as in figure 1.13. Here we emphasize that actions are implemented periodically, leading to system

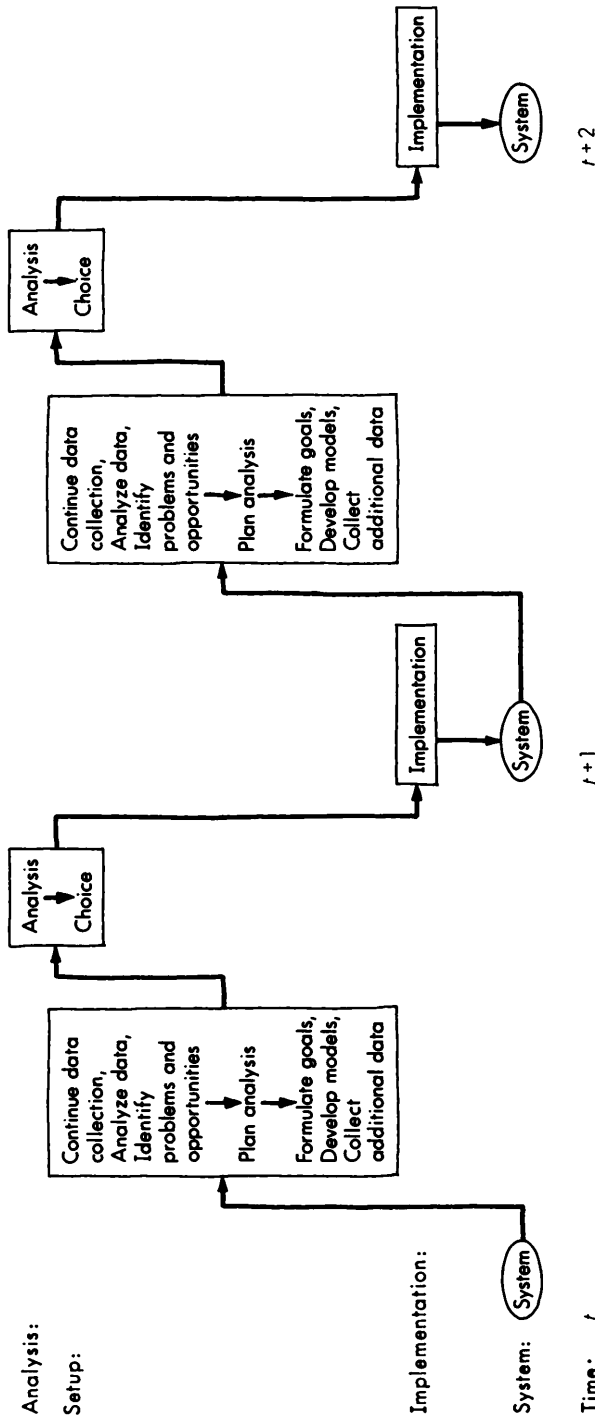


Figure 1.13 The continuing analysis process—Image 2.

changes. The system evolves over time, influenced partly by internal dynamics (the interactions of transportation and activity systems) and partly by actions taken deliberately as a result of analysis.

The synthesis of the images of figures 1.12 and 1.13 can be described in terms of the product of analysis, the ***multiyear program plan*** (MYPP) shown in figure 1.14. The MYPP reflects the view that the product of analysis is a program of actions staged over an extended period, which might range up to perhaps twenty-five years. The outcome of the first period of the plan is firm and detailed: it is the set of specific implementable actions that will be taken in the next period. That is, inclusion of an action in the first period of the MYPP represents a decision to implement that action in the next period. That first period may be a year in the case of a regional or national planning agency, or a quarter (three months) in the case of an operator such as a transit agency or airline. Each succeeding period of the plan is less firm and less detailed. The actions in the first few periods (up to year three or five) of the MYPP constitute the "Short-Range Plan"; those further off (say, in years six to fifteen), the "Midrange Plan"; and later actions (say, in years sixteen to twenty-five) the "Long-Range Plan." These time periods may be shorter or longer in different organizational contexts. For example, for an airline the short range may be three to six months, the midrange six to twenty-four months, and the long range two to five years.

The MYPP contains all significant transportation actions proposed for the system. For a region these would cover all modes and all types of transportation and related options: changes in facilities, in vehicle fleets, in operating policies (routes, schedules, fares, classes of services, restraint and other disincentive policies), in organizations and institutions, and in transportation-related actions (staggered work hours, land-use controls, sewer and water policies). For a private carrier such as a trucker or an airline the actions would be more restricted, such as changes in vehicle fleets, routes, schedules, and services.

The MYPP also includes a listing of studies to be undertaken. These actions are as important as implementation actions because they influence, and sometimes constrain, the implementation actions that can be seriously considered in future years—you cannot consider implementing a major new service strategy *next* year if you didn't do some homework on it *this* year.

The MYPP logically should have different degrees of detail for different types of actions. The MYPP for a region such as a metropolitan area or state may contain component MYPPs for various sub-

Action	Year										
	1 Implement	2	3	4	5	6	...	15	15	...	25 or 30
		Short range				Midrange			Long range		
Implementation											
Facilities: New Improvements											
Vehicles											
System operating policies (including pricing)											
Organization											
Transportation- related activity- system actions											
Studies											
Planning											
Design											
Monitoring											
Evaluation											
Research											
Costs											
Category A											
•											
•											
•											
Category Z											

Figure 1.14 The transportation multiyear program plan (Manheim 1977a).

regions. Different levels of detail would be appropriate for different levels of geographic aggregation. The MYPP for a county or medium-sized city would be more detailed than that for a state. The state MYPP would summarize in a single item what might be a large number of specific actions in a county MYPP. Similarly, the MYPP for an operator's system may contain MYPPs for portions of the system.

A sample MYPP for a particular region is given in figure 1.15.

Where decisions on actions to be taken in future years have not been made yet, or are contingent on alternative outcomes of earlier actions, the MYPP can show these. For example, unresolved issues can be included or explicit contingencies shown, as in figure 1.15.

The MYPP must be reviewed periodically and revised according to an explicit decision process in which progress in implementing actions in the first period of the preceding year's MYPP is reviewed. Obviously, not all elements of the MYPP would be subject to the same degree of scrutiny in each review cycle. For example, the long-range portion of the MYPP might be revised in a major way only once every three or five years, or when a decision is to be taken on implementing a major project in the next year.

In the periodic decision process, some of the actions in the second year of the preceding year's MYPP may be advanced into the first year of the new MYPP, reflecting decisions to implement those actions; others may be deferred for implementation or may be discarded altogether. This reflects the fact that while the MYPP lays out planned future actions, in actuality the only firm decision is an implementable one: how to spend next year's dollars. Since actions indicated in the MYPP are never certain of implementation until they move into the first year of a current MYPP, an important issue in choosing an action for implementation in a given year is its degree of "commitment" versus "flexibility": If this action is implemented, which future options will be foreclosed, and which will remain open for future implementation?

1.6.4 Implications for Analysis

Designing an analysis thus involves more than just developing a prediction model. The analyst must develop a ***strategy for analysis*** that includes plans for the full range of necessary technical activities:

1. What data should be collected?
2. What models should be developed?
3. How should the tasks of search, prediction, and evaluation and choice be organized into a coherent process?

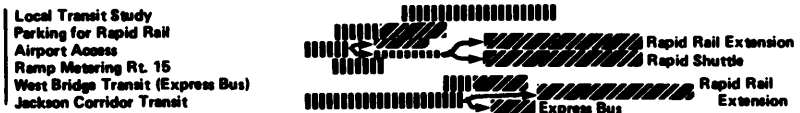
PROGRAM PACKAGE FORMAT

Project No. | Major Improvement By Corridor

NORTH CORRIDOR



CENTRAL CORRIDOR



SOUTH CORRIDOR



REGIONAL POLICY CHANGES



COST: \$ IN MILLION	Study	10	28	57
	Highway Implementation	19	18	36
	Highway Operating	-	-	-
	Transit Implementation	72	239	477
	Transit Operating	300	520	1077
	TOTAL	401	805	1647



0 5 10 20

Figure 1.15 A sample multiyear program plan (Manheim et al. 1975b, based on Neumann 1976).

4. How should technical activities interface with the political process?
5. What should be the nature of the continuing analysis and implementation process?

1.6.5 Systematic Analysis

The general structure of a systematic analysis is illustrated metaphorically in figure 1.16: the options T are varied systematically over a range, and the impacts on various interests—here, users, operators, and government—are traced out as T is varied. For example, it is often especially important to explore trade-offs among various types of options—increasing frequency of service versus reducing fares to increase ridership; changing the relative mix of resources expended on new facilities as opposed to operating expenses and improvements in existing facilities; or changing the relative mix of expenditures on various modes.

Of course, the prediction of impacts for each T requires use of the system of prediction models. Practically, therefore, the metaphor may be difficult to apply in some situations. As we shall demonstrate in later chapters, however, a systematic analysis can be achieved in most situations. Practical strategies involve careful design of the iterations of the basic analysis cycle of search-prediction-evaluation to trace out trade-offs.

1.6.6 Analysis and the Political Process

To intervene effectively in the fabric of society and bring about change through the implementation of a transportation system plan, the analyst

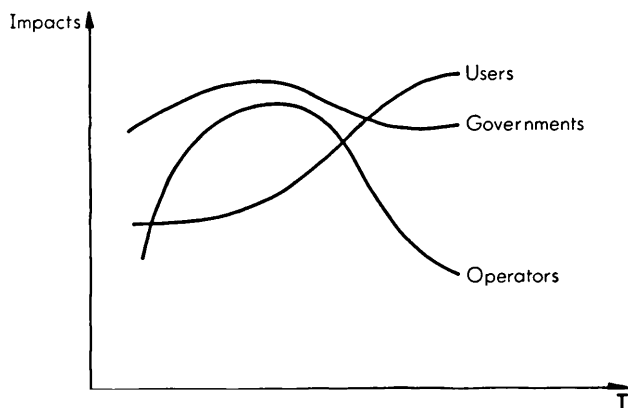


Figure 1.16 The general structure of a systematic analysis.

must have a perspective on the relation between analysis and the political process.

The transportation systems analyst usually operates in a complex institutional environment, whether public or private. In such an environment there are usually many different groups that will have an interest in the outcome of the analyst's work, and they will all have different ideas about what it is desirable to achieve with transportation. Typically, also, there are many different institutions involved.

To be useful, analysis must be relevant to the issues with which these institutions and interest groups are concerned (or are likely to be concerned), and must clarify the issues in ways that contribute to the constructive resolution of potential conflicts and to the making of implementable decisions by responsible decision makers.

With this view, and with the definition of systematic analysis in the preceding section, we see that our definition of the methodological challenge of transportation systems analysis (section 1.5.7) needs to be broadened, by adding the concepts of "systematic analysis" and "clarification of issues": ***The methodological challenge of transportation systems analysis is to conduct a systematic analysis in a particular situation which is valid, practical, and relevant, and which assists in clarifying the issues to be debated.***

In responding to this challenge, the analyst must be concerned with many basically different kinds of activities. Our focus in this book will be primarily on prediction, with some attention to evaluation. As a result of this concentration, we shall develop a rather abstract, technical view of transportation systems analysis. This is necessary in order to foster the understanding of fundamentals that is essential to doing good analysis. But it must be kept in mind that transportation systems analysis must be rooted in the complexities of social and political reality. Ultimately we must return from the delights of abstract analytics to the realities of social and political issues and forces.

Successful response to the challenge requires deep understanding, not only of the theory and techniques of prediction, but also of search, evaluation, choice, the structuring of a process of analysis, and, most importantly, the social, economic, and political context in which an analysis is being conducted.

1.7 SUMMARY

We live in a world of rapid change—change in transportation technology, in transportation demand, and in the values, public and private,

that will influence decisions about transportation systems. The challenge of transportation systems analysis is to use the transportation system to help a society achieve its goals. Therefore, analysis must be a continuous process. Setup, technical analysis, and implementation must be continuously iterated and adjusted to changing conditions, with the adopted course of action revised appropriately.

At the core of this process is the prediction task: based upon the current understanding of the mechanisms of interaction between transportation and activity systems, a set of models is developed and used to predict the impacts of proposed courses of action. The heart of this prediction of impacts is the prediction of flows in networks, based upon service–demand equilibrium concepts. Although simple in concept, prediction is a major technical task.

It is this task of prediction to which we turn first in the following chapters. Though the discourse will become highly technical and somewhat abstract, it is important that the reader keep in mind the broader context in which these techniques are embedded.

TO READ FURTHER

For further discussion of the interrelation of transportation and activity systems, see S. B. Warner (1962) and Meyer, Kain, and Wohl (1965). The best sources of information on current transportation issues and news are newspapers, news magazines, and industry-oriented publications such as *Railway Age*, *U.S. Transport*, *Mass Transit*, and *Aviation Week and Space Technology*.

EXERCISES

1.1(C) Select a current transportation issue that interests you and that involves the transportation services of a particular region or operator. Identify the key elements you think should be included in an analysis of this issue, such as:

- a. major options to be analyzed
- b. major impacts to be predicted
- c. major flow types
- d. service attributes
- e. prediction procedures
- f. possible search procedures
- g. what the choice process might be like.

1.2(P) Select a transportation study for which you have adequate documentation (a final report, environmental impact statement, or an internal staff study).

a. Summarize concisely the key issues that the study seems designed to address.

b. Identify the key elements of the study:

i. system considered (refer to the system principles in section 1.2)

ii. options and impacts considered

iii. flow types considered (passengers and/or commodities)

iv. service attributes considered

v. prediction procedures used (demand functions, activity-shift functions, service and resource functions, travel-market equilibration, operator and activity-system equilibration procedures)

vi. search procedures and evaluation procedures

vii. choice process

viii. assumptions about exogenous events.

c. Discuss the study elements:

i. Does the system definition seem appropriate to the issues? If not, why not?

ii. Do the options and impacts seem appropriate? If not, why not?

iii. Continue the discussion with other elements identified in part b.

1.3(P) Choose two of the following contexts: air carrier, trucking company, shipping company, port authority, national highway agency.

a. What are the major types of options that might be considered by this organization?

b. Sketch out a hypothetical multiyear program plan for this organization.

c. What would be the basic features of the prediction models required to predict the impacts of this MYPP?

1.4(E) In railroad operations there is substantial debate about train length. In a particular market longer trains reduce crew costs. On the other hand, by operating more frequent, shorter trains, "railroads incur greater crew costs, but obtain better service and improved freight car utilization. Furthermore, short trains can be handled more easily at yards, at sidings, and on the line than long trains. In some instances, however, running more short trains could cause operating delays on congested portions of the system" (Martland, Assarabowski, and McCarren 1977). Consider the situation of a 200-mile segment of rail line on which the present operating policy is to run two trains per day with 90 cars each, or a total capacity of 180 cars per day:

$$\mathbf{T}^0 = (Q^0, L^0) = (2, 90),$$

where

Q = frequency (trains per day),

L = train length (cars per train).

The cost per train in dollars is (after Martland 1977)

$$C_{\text{TRAIN}}(Q, L) = 2.40 \frac{L}{Q} + (480 + 1.33L + 0.004L^2)$$

or, for a day's schedule of Q trains,

$$C_{\text{DAY}}(Q, L) = QC_{\text{TRAIN}} = 2.40L + Q(480 + 1.33L + 0.004L^2).$$

The total capacity available in cars per day, V_C , is

$$V_C = QL.$$

Operating management believes that the following demand function applies in this corridor (V_D is demand volume):

$$V_D(Q, L) = 180 \left(\frac{Q}{Q_0} \right)^\beta;$$

a "best estimate" of β is 0.5.

Management's objective is to increase net revenue, I_{NR} , where

$$I_{\text{NR}} = I_{\text{GR}} - C_{\text{DAY}}$$

and gross revenue, I_{GR} , is a function of equilibrium volume V_E and price P :

$$I_{\text{GR}} = V_E P.$$

At present $P = \$15$ per carload for freight over this segment.

a. Consider the possibility of adding 1, 2, or 3 more trains per day, maintaining the same train lengths. Predict the equilibrium flows and the corresponding costs and revenues for these strategies. Which of these strategies (T^0, T^1, T^2, T^3) would you recommend to management, and why?

b. If your objective were maximum service to users, which would you recommend, and why? Compare with your answer to part a and discuss.

c. For a given frequency, what are the advantages and disadvantages of varying train lengths? If you knew that for a frequency Q^i the demand would be V^i , what train length would you recommend?

d. Consider these alternative strategies: $T^4 = (Q^4 = 3, L^4 = 75)$, $T^5 = (4, 75)$, $T^6 = (5, 75)$, $T^7 = (4, 70)$, $T^8 = (5, 70)$, $T^9 = (5, 60)$. Which of these would you recommend from the operator's perspective? for $Q = 4$? for $Q = 5$? Which of the ten strategies T^0 – T^9 would you

recommend from the operator's perspective? from the user's perspective? Discuss.

e. Using your result to part c, find the train length L_{OPT} that is best from the operator's perspective for each frequency $Q = 3, 4, 5$, and predict the impacts of the corresponding strategies. Plot the results on axes defined by V_D and I_{NR} and discuss. Which would be best from the operator's perspective? from the user's perspective? Why? Which would you recommend, and why?

f. How might the results change for $\beta = 0.2$? $\beta = 0.8$? (Check only for $L = L_{\text{OPT}}$ for this range of frequencies.)

g. Critique the assumptions (implicit and explicit) of this formulation of the problem: Under what conditions, and to what extent, do you think they would be reasonable? (For example, what do you think of the explicit assumption that demand is independent of the train length?)

1.5(E) The country of Freelandia gained independence a few years ago and is mounting a major effort to promote new agricultural development in previously underdeveloped regions. A trucking operator in the town of K has previously been providing only local service. Now that a new major agricultural development program is under way, this operator is considering providing farm-to-market service to carry agricultural and other natural products from their origin in locality M to market at K . The distance is 150 miles (one way), with no intermediate major settlements.

After discussions with the local agents of the producers at M , the trucker estimates that the demand function for shipments from M to K is

$$V = Z + a_0Q - a_1P,$$

where V is volume in tons per week, Q is frequency of shipments (per week), P is price charged per ton, and a_0 , a_1 , and Z are parameters. Based upon an average traveling speed of 30 miles per hour, plus a loading or unloading time of 3 hours at each end, he estimates that he can manage at the most one round trip every two days, so $Q = 3$ per week. He also figures that his costs are related to the mileage he drives per year: his total cost per year is

$$C_T = b_0 + b_1m_T,$$

where $m_T = 300Q$ is the total round-trip mileage driven and b_0 and b_1 are parameters. The truck carries 15 tons. He is considering offering an initial frequency of 1 or 2 trips per week at a rate of \$25.00 or

\$30.00 per ton. Assume $b_0 = \$270$, $b_1 = \$0.50$, $Z = 25$, $a_0 = 13$, $a_1 = 1$.

a. For these four combinations of frequency and price, what would be the tonnage carried, the gross revenues, the total cost, and the net revenue?

b. Which of the four options would be preferred by the operator if his objective were to maximize net revenue? to minimize costs? to maximize volume carried? Which option would be preferred by users (shippers)? Can both interests get their first choice simultaneously? If not, why not?

c. For the proposed service the predominant movement is from M to K ; the amount of freight to be carried in the reverse direction is negligible. There is a possibility of picking additional cargo at D to go to M ; this would incur a detour of 100 miles additional but could result in an additional load and source of revenue. Would it be profitable for this operator to make the detour? Discuss qualitatively.

1.6(E) A railroad operator utilizes a fleet of 2,000 freight cars for service to a particular group of shippers, who produce grain. At present train schedules are poorly coordinated. Because service is unreliable, shippers take an excessive amount of time to load and unload the cars, with the result that an average car delivers $n = 10$ carloads per year (that is, the cycle time is $t_c = 365/10 = 36.5$ days).

The cost of owning a railroad car is \$2,000 per year (equivalent annual cost). At present shippers are charged $p = \$1,200$ per load, and the operating costs (excluding ownership) are $a = \$1,100$ per load.

The service now runs at a loss. Management is considering three options: (A) increase rates; (B) improve schedule coordination and other aspects of operation so that service is more reliable, delivery times are shortened, and shippers load and unload cars more quickly; or (C) do both of the above. The marketing staff has come up with the following approximate demand function showing the effect on total carloads V of a change in rates, in cycle time, or in both:

$$\frac{\Delta V}{V_0} = \alpha \frac{\Delta p}{p_0} + \beta \frac{\Delta t_c}{t_{c0}}.$$

a. Assume $\alpha = -0.4$, $\beta = -1.2$, $V_0 = 2,000 \times 10 = 20,000$. Predict the new volume V' for (A) $\Delta p = +\$100$; (B) $\Delta t_c = -4.5$ days; and (C) both.

b. Assuming that the fleet size stays the same, predict for each option the increase in annual revenues from shippers (Vp), the increase in

operating costs ($2,000na$), and the change in net operating revenue ($Vp - 2,000na$). Which option would you recommend?

c. Which option would you recommend if the cost of implementing option B turned out to be equivalent to an increase in operating costs to $a = \$1,200$ per load?

d. Do a sensitivity analysis for the assumed demand parameters, using $\alpha = -0.8$ and -1.2 ; $\beta = -0.4$ and -0.8 . Discuss.

1.7(E) A marine operator presently offers service between ports A and B , 6,000 miles apart, utilizing a single ship. At an equivalent land speed of 20 miles per hour (ship speeds are usually expressed in nautical miles or knots), the total sailing time for a one-way voyage is 300 hours or 12.5 days. The average time spent loading or unloading cargo in either port is two days, so the average one-way trip time for the ocean voyage for a cargo is $12.5 + 0.5(2 + 2) = 14.5$ days. The round-trip time is thus 29 days. For a "working year" of 330 days (allowing time for periodic maintenance), the effective frequency is $330/29 = 11.4$ round trips per year. The ship's capacity is 15,000 tons of cargo; the rate presently charged is \$25 per ton. The average cost of a one-way trip is \$225,000 per voyage, or \$15 per ton of available capacity.

The operator estimates that the demand function in this market is

$$V = Z_0 - \alpha t_{iv} - \beta/Q - \gamma c,$$

where

V = round-trip volume in tons per year

Z_0 = market size factor

t_{iv} = one-way trip time

Q = frequency in round trips per year

c = freight rate in dollars per ton

α, β, γ = parameters.

His estimates of the parameters are $\alpha = 19,500$, $\beta = 3.1 \times 10^6$, $\gamma = 8,000$, and $Z_0 = 982,680$. The average volume per voyage is 20,000 tons.

a. Are his estimates of the parameters consistent with this volume? What fraction of available capacity is utilized per voyage? What is his gross revenue (receipts from rates paid), total cost, and net revenue (gross revenue less total cost) per voyage?

b. The operator is considering replacing the present ship with a newer vessel, which would have a speed of 24 miles per hour, thus

cutting the one-way sailing time to 10.4 days. The cost per voyage of this newer, faster ship would be \$250,000 per one-way trip. The capacity and loading/unloading times would be the same. Would this ship be more attractive to the operator: (i) if frequency and rate remained the same? (ii) if the frequency were increased to take advantage of the increased speed but the rate remained the same? (iii) if the frequency were increased and the rate increased to \$30.00 per ton? Summarize and discuss your results: What consequences can a change in vehicle speed have? Discuss the significance of the parameters α , β , and γ .

c. After doing this analysis, the operator suddenly realizes that he has ignored a fundamental principle: the time that will influence shipper's decisions (that is, the demand function) should be the total door-to-door time, not just the trip time on the ocean leg. He estimates that an average shipment spends about six days in the land portion of the trip, for a total travel time of 20.5 days. How might this affect his estimates? (Discuss qualitatively.)

1.8(E) A railroad runs from A to B , a distance of 500 miles, through mountainous terrain. The present one-way travel time (including time at intermediate yards) is 20 hours, and the rail freight rate is \$20 per ton. There is a truck service which competes with the railroad, running over roughly parallel roads for approximately the same distance, at an average speed of 30 miles per hour and at a rate of \$30 per ton. A new highway is planned to replace the existing road; while there is some auto traffic, it is expected that most of the traffic will be trucks. The service function of the new facility is $t_T = t_0 + bV_T$, where V_T is the total volume in trucks per hour (the anticipated auto usage is negligible), $t_0 = 10$ hours, $b = 0.08$ hour per truck per hour. The railroad's estimate of the demand function is

$$\frac{V_T}{V_R} = a_0 \left(\frac{t_T}{t_R} \right)^{a_1} \left(\frac{c_T}{c_R} \right)^{a_2},$$

where t_T and t_R are the trip times by truck and rail, respectively, c_T and c_R are the rates, V_T and V_R are the volumes, and a_0 , a_1 , and a_2 are parameters. The total volume is likely to remain constant at $V_{TOT} = V_T + V_R = 200$ truckloads per hour. The rail system is utilized at only a fraction of capacity, so its service function is flat—travel time is constant independent of volume.

a. If $a_0 = 1$, $a_1 = -1$, and $a_2 = -2$, find the present volumes (mode split) of truck and rail.

b. Make an approximate estimate of the equilibrium flows if the new highway were built. (*Hint:* Try a range of truck volumes in a trial-and-error approach.)

c. What would the equilibrium flow be if the railroad dropped its rates to \$15 per ton? if truckers were taxed \$5 per ton to help pay for the new highway? if both changes occurred? Discuss.

d. Discuss the advantages and disadvantages of the new highway from the perspectives of each of the major affected interests.

e. A regulatory commission is examining the question of rates. The railroad is arguing for a truck rate of \$35 per truckload, while the truckers are arguing for a rate of \$30. What would be the corresponding equilibrium volumes? Why is each interest advocating the particular rates indicated?

1.9(E) An urban expressway presently carries a peak traffic volume of 2,200 vehicles per hour over three lanes in the peak direction. A simple, approximate service function for this facility is

$$t = t_0 + b \frac{q}{kq_c},$$

where k is the number of lanes, $q_c = 1,200$ vehicles per hour per lane, q is the total one-way flow volume in vehicles per hour ($q \leq 0.95kq_c$), $b = 3$, and $t_0 = 2$ minutes per mile (see section 7.6). One bus is considered equivalent to about 1.6 automobiles, and the flow stream in the peak hour includes about 60 buses. The demand function for bus transit usage in the corridor is

$$V_B = V_0 - at_B,$$

where t_B is bus travel time in minutes, $V_0 = 4,200$ people per hour, and $a = 75$.

a. What is the present travel time for buses and automobiles over the eight-mile expressway? Show that the ridership on the 60 buses is $V_B = 1,982$ persons per hour.

b. It is proposed that one auto lane be replaced by a lane for exclusive use of buses, with semipermanent barriers such as rubber cones between the bus and auto lanes. If the number of buses in the peak hour remained the same, what would their travel time be? (Assume no en route stops and no congestion for buses at exit ramps.) What would be the equilibrium volume of passengers using buses? If the maximum capacity of the buses were 50 passengers, would they be crowded?

- c. How would the automobile travel times change if one lane were set aside for the exclusive use of buses?
- d. What assumption is made in this model about the influence of automobile travel times on bus ridership? If you relaxed this assumption, how might the results change?